Variable Description

Variable	Description
patient_id	Unique identifier assigned to each patient record
gender	Gender of the patient
age	Age of the patient in years
age_group	Age categorized into groups such as young, middle-aged, and senior
has_hypertension	Indicates whether the patient has hypertension (1 = Yes, 0 = No)
has_heart_disease	Indicates whether the patient has heart disease (1 = Yes, 0 = No)
marital_status	Marital status of the patient (1 = Married, 0 = Not married)
employment_type	Type of employment or work status of the patient
residence	Type of residence area (Urban or Rural)
glucose_level	Blood glucose level of the patient
high_glucose	Indicates whether glucose level is considered high (1 = Yes, 0 = No)
bmi_value	Body Mass Index value of the patient
bmi_category	BMI classification such as underweight, normal, overweight, or obese
smoking_habit	Smoking status of the patient (e.g., non-smoker, ex-smoker, unknown)
lifestyle_risk	Overall lifestyle risk category (e.g., low, medium, high)
risk_score	Composite score representing overall health risk based on multiple factors
stroke_event	Indicates whether the patient experienced a stroke (1 = Yes, 0 = No)

Case Study

Predicting Stroke Risk Using Patient Health and Lifestyle Data

Background

Stroke remains one of the most critical global health challenges, contributing significantly to mortality and long-term disability. Early identification of individuals at high risk can enable timely intervention, reduce healthcare costs, and improve patient outcomes.

Modern healthcare systems collect large volumes of patient data, including demographic details, clinical measurements, and lifestyle information. However, extracting meaningful insights from this data to support decision-making remains a challenge.

Scenario

You are part of a data science team working with a healthcare organization that aims to improve early detection of stroke risk among patients.

You are provided with a dataset containing **9,722 patient records**, including information on:

- Demographics
- Medical history
- Lifestyle habits
- Clinical measurements

The dataset also includes whether a patient has experienced a stroke.

Problem Statement

Using the available dataset, your task is to:

- Develop a model to **predict the likelihood of a stroke event**
- Identify the **most influential factors contributing to stroke risk**
- Generate **insights that can support preventive healthcare strategies**

Key Tasks

1. Data Understanding & Exploration

2. Data Preparation and Visualization

3. Model Development

4. Insight Generation